AT1-Stage-1- Story Artifact | Week-3 Reflective Journal #1

~~Rohith Syam [25754704] , Moniz [25614049], Trishna [26521474] , Shuvam [26619344], Jacqueline Chan [13455074] , Muhammad Saad [26212427]~~

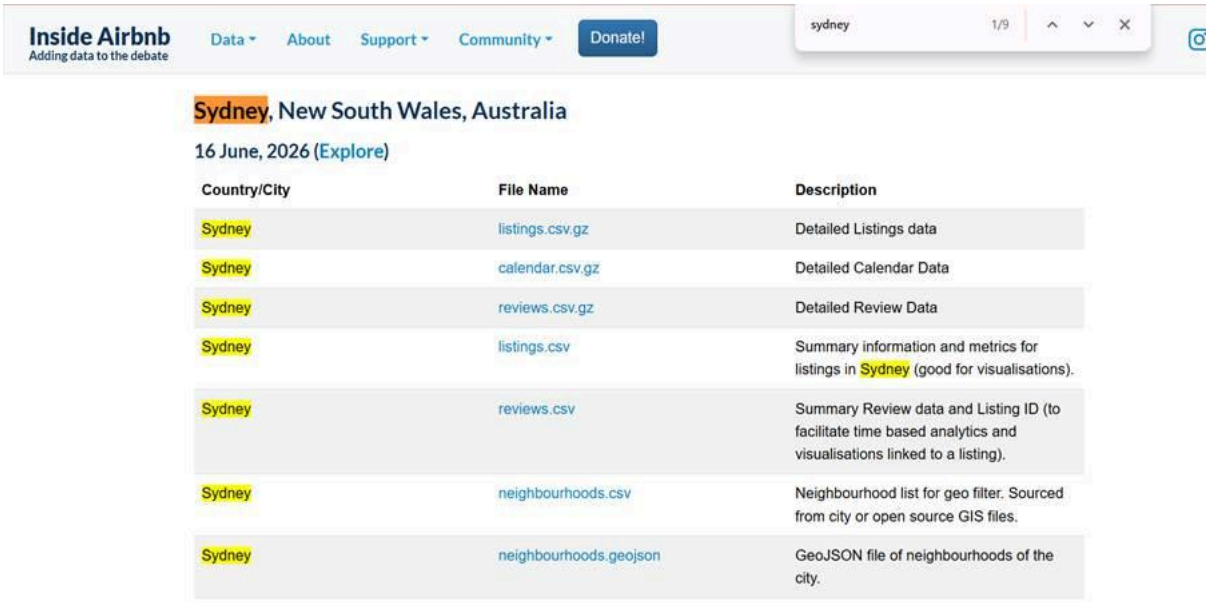
Exploring Sydney's Airbnb Market: Housing Availability and STRA Regulation

Colab link referenced throughout the document: airbnb_sydney_preprocessing.ipynb

1. Topic and Dataset Selection

Our project explores Sydney's Airbnb market, focusing on how short-term rental accommodation may relate to long-term housing availability and NSW STRA regulation. This is particularly relevant because Sydney is experiencing significant rental affordability pressure. The City of Sydney reported a rental vacancy rate of approximately 1.5 per cent, indicating limited availability in the long-term rental market (City of Sydney, 2025). Research also suggests that high concentrations of STRA can place additional pressure on rents and housing supply by converting properties from long-term to short-term accommodation (Gurran et al., 2026). The cleaned dataset also contains 1,269 listings, around 7 per cent of the dataset, with no reported licence information, creating an important area for further investigation rather than evidence of non-compliance. By combining reported licence status with room type, host scale, annual availability and location, our project provides a more targeted way to identify listings and areas that may warrant further review.

We used Inside Airbnb's detailed Sydney listings dataset dated 16 June 2026 (Inside Airbnb, 2026). The raw dataset contained 20,573 listings and 90 columns, including information about prices, availability, host activity and reported licence details.



Inside Airbnb
Adding data to the debate

Data ▾ About Support ▾ Community ▾ Donate!

sydney 1/9

Sydney, New South Wales, Australia

16 June, 2026 (Explore)

Country/City	File Name	Description
Sydney	listings.csv.gz	Detailed Listings data
Sydney	calendar.csv.gz	Detailed Calendar Data
Sydney	reviews.csv.gz	Detailed Review Data
Sydney	listings.csv	Summary information and metrics for listings in Sydney (good for visualisations).
Sydney	reviews.csv	Summary Review data and Listing ID (to facilitate time based analytics and visualisations linked to a listing).
Sydney	neighbourhoods.csv	Neighbourhood list for geo filter. Sourced from city or open source GIS files.
Sydney	neighbourhoods.geojson	GeoJSON file of neighbourhoods of the city.

Figure 1: Inside Airbnb Sydney data source (16 June 2026).

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1	id	listing_url	scrape_id	last_scrap	source	name	description	neighborh	picture_url	host_id	host_url	host_profil	host_profil	host_nam	host_since	hosts_time	hosts_time	hosts_time	hosts_time
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Figure 2: Raw dataset dimensions - 20,573 listings × 90 columns.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20573 entries, 0 to 20572
Data columns (total 90 columns):
 #   Column                                        Non-Null Count  Dtype
---  ------                                        --------------  -----
 0   id                                            20573 non-null  int64
 1   listing_url                                   20573 non-null  object
 2   scrape_id                                     20573 non-null  int64
 3   last_scraped                                  20573 non-null  object
 4   source                                        20573 non-null  object
 5   name                                          20573 non-null  object
 6   description                                   20243 non-null  object
 7   neighborhood_overview                         0 non-null      float64
 8   picture_url                                   20573 non-null  object
 9   host_id                                       20573 non-null  int64
 10  host_url                                      20573 non-null  object
 11  host_profile_id                               20573 non-null  int64
 12  host_profile_url                              20573 non-null  object
 13  host_name                                     20573 non-null  object
 14  host_since                                    0 non-null      float64
 15  hosts_time_as_user_years                      20573 non-null  int64
 16  hosts_time_as_user_months                     20573 non-null  int64
 17  hosts_time_as_host_years                      20573 non-null  int64
 18  hosts_time_as_host_months                     20573 non-null  int64
 19  host_location                                 14750 non-null  object
...
 88  calculated_host_listings_count_shared_rooms   20573 non-null  int64
 89  reviews_per_month                             17247 non-null  float64
dtypes: float64(37), int64(24), object(29)
memory usage: 14.1+ MB
```

```
keep = [
    'id', 'name', 'host_id', 'host_name', 'host_is_superhost',
    'host_has_profile_pic', 'host_identity_verified',
    'neighbourhood_cleansed', 'latitude', 'longitude',
    'property_type', 'room_type', 'accommodates', 'bathrooms_text',
    'bedrooms', 'beds', 'amenities',
    'price', 'minimum_nights', 'maximum_nights',
    'has_availability', 'availability_30', 'availability_60',
    'availability_90', 'availability_365',
    'number_of_reviews', 'number_of_reviews_ltm', 'number_of_reviews_l30d',
    'first_review', 'last_review',
    'review_scores_rating', 'review_scores_accuracy', 'review_scores_cleanliness',
    'review_scores_checkin', 'review_scores_communication',
    'review_scores_location', 'review_scores_value',
    'license',
    'estimated_occupancy_l365d', 'estimated_revenue_l365d', 'reviews_per_month',
    'calculated_host_listings_count',
    'calculated_host_listings_count_entire_homes',
    'calculated_host_listings_count_private_rooms',
    'calculated_host_listings_count_shared_rooms',
]
keep = [c for c in keep if c in df.columns]
print(f'{len(keep)} columns kept, {df.shape[1] - len(keep)} dropped')
df = df[keep].copy()

45 columns kept, 32 dropped
```

Fig 3: Initial exploration of the raw dataset. Fig 4: Retained fields after removing unusable columns



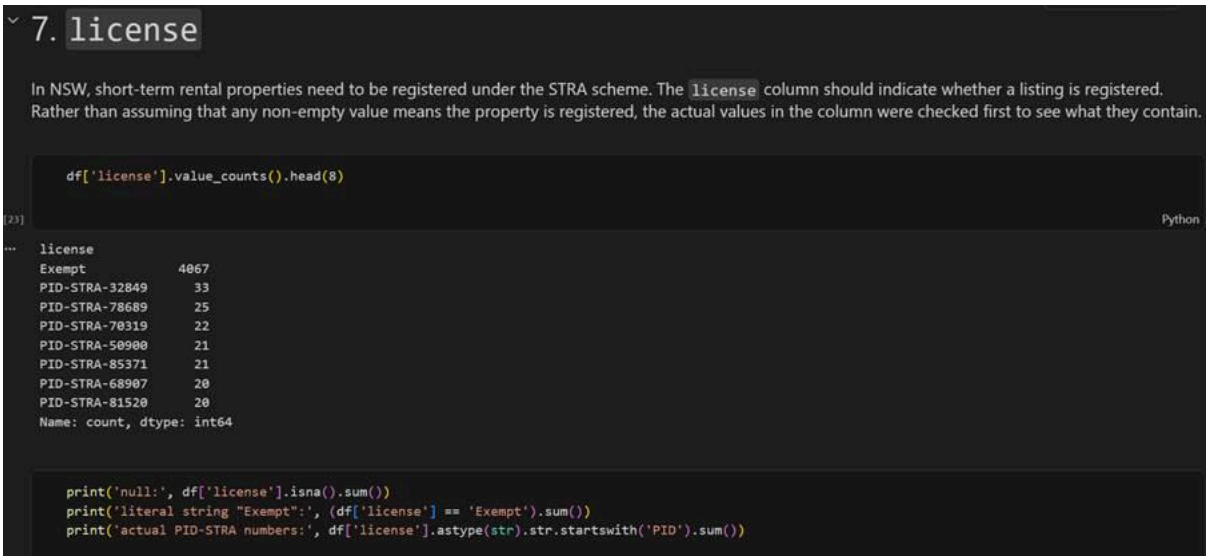
4. bathrooms_text

No numeric `bathrooms` column in this scrape — the count lives in this text field instead. Checking actual values before writing a parser.

```
df['bathrooms_text'].value_counts().head(15)
```

```
bathrooms_text
1 bath              9430
2 baths             4245
1 shared bath       1479
1 private bath      1337
3 baths              863
1.5 baths            698
2.5 baths            690
2 shared baths       453
1.5 shared baths     298
3.5 baths            245
4 baths              220
0 shared baths        95
4.5 baths             88
0 baths               76
5 baths               71
Name: count, dtype: int64
```

Figure 5: Decoding bathrooms_text.



7. license

In NSW, short-term rental properties need to be registered under the STRA scheme. The `license` column should indicate whether a listing is registered. Rather than assuming that any non-empty value means the property is registered, the actual values in the column were checked first to see what they contain.

```
df['license'].value_counts().head(8)
```

Python

```
license
Exempt           4067
PID-STRA-32849     33
PID-STRA-78689     25
PID-STRA-70319     22
PID-STRA-50900     21
PID-STRA-85371     21
PID-STRA-68907     20
PID-STRA-81520     20
Name: count, dtype: int64
```

```
print('null:', df['license'].isna().sum())
print('literal string "Exempt":', (df['license'] == 'Exempt').sum())
print('actual PID-STRA numbers:', df['license'].astype(str).str.startswith('PID').sum())
```

Figure 6: Licence field inspection and PID-STRA values.



Figure 7: Licence categorisation logic: registered (PID-STRA), self-declared exempt, and no license info.

Colab link : airbnb_sydney_preprocessing.ipynb

- ## Results After Preprocessing and Cleaning

Preprocessing produced a smaller and more focused dataset for analysing Sydney's Airbnb market.

Dataset	Listings	Variables
Original dataset	20,573	90
Final dataset	17,741	55

The final dataset includes listing and host characteristics, location, property and room types, prices, minimum-stay requirements, availability, reviews, amenities, reported licence information, host activity, and estimated occupancy and revenue. Together, these variables support the analysis of Sydney's Airbnb market and reported STRA activity.

The complete list of retained variables and preprocessing steps is available in the Colab notebook: airbnb_sydney_preprocessing.ipynb

- ## Improved for Storytelling:

Several variables were created to make the original fields easier to interpret and use in the data story.

New variable	How it was created	Value for the story

bathrooms, bathrooms_shared	Derived from the free-text bathrooms_text field. For example, “2 shared baths” became bathrooms = 2 and bathrooms_shared = True. Half-baths were recorded as 0.5.	Allows bathroom information to be compared consistently across listings.
amenities_count	Counts the amenities reported by each listing after the original text was converted into a Python list.	Provides a simple measure of how well-equipped a listing is.
has_pool, has_aircon, has_parking, has_wifi	Created as Boolean indicators from the amenities list.	Supports investigation of potential differences in listing prices and demand.
license_status	Groups the original license field into registered (PID-STRA), self-declared exempt and no licence information.	Makes reported STRA licence information easier to compare.
stra_registered	Set to True when a listing displays a PID-STRA-formatted identifier.	Supports analysis of reported STRA registration. It does not independently verify the identifier against the NSW STRA Register.
is_multi_listing_host	Set to True when calculated_host_listings_count is greater than one.	Helps identify hosts who appear to operate multiple listings.

- ## End-to-End Airbnb Data Preprocessing Pipeline:



The Sydney listings dataset was downloaded from Inside Airbnb. The dataset, dated 16 June 2026, originally contained 20,573 listings and 90 columns. The main preparation steps are summarised below.

Stage	**What was done**

Selecting relevant fields	Thirteen empty columns were removed, along with unnecessary free-text fields, URLs, and scrape-related metadata. This left 45 relevant original fields.
Converting data	Price was converted to a numeric field, bathrooms_text was separated into structured variables, t and f values were converted to Boolean values, and amenities was converted into a Python list using ast.literal_eval.
Creating new variables	New variables were created for bathrooms, amenity counts, selected amenities, reported licence status, multi-listing hosts, and the exploratory 180-day availability indicator.
Filtering listings	Listings with missing prices, prices below $10 or at least $10,000, and minimum stays above 365 nights were removed.
Renaming fields	neighbourhood_cleansed was renamed lga to make the field easier to understand.

Listings with missing prices had a median availability_365 of zero. This suggests that many may have been inactive or unavailable when the data was collected, although the dataset alone cannot confirm this.

After cleaning and feature engineering, the final dataset contained 17,741 listings and 55 columns.

- # Value Proposition of the Data-Driven Story

Value Proposition

This project examines whether Airbnb is drawing homes away from Sydney's long-term rental market and using them for short-term stays instead. By analysing property type, availability, host activity, registration status, and location, we can distinguish genuine home sharing from likely commercial short-term rentals. This could help NSW regulators and councils flag possible compliance issues.

Significance

Sydney faces ongoing rental affordability and availability pressures. Of 17,741 active Airbnb listings, 13,805 (78%) are entire homes rather than shared rooms, and 48% of these are available over 180 days a year. This suggests many properties that could support long term tenancy are operating as short-term rentals, adding pressure to a tight rental supply.

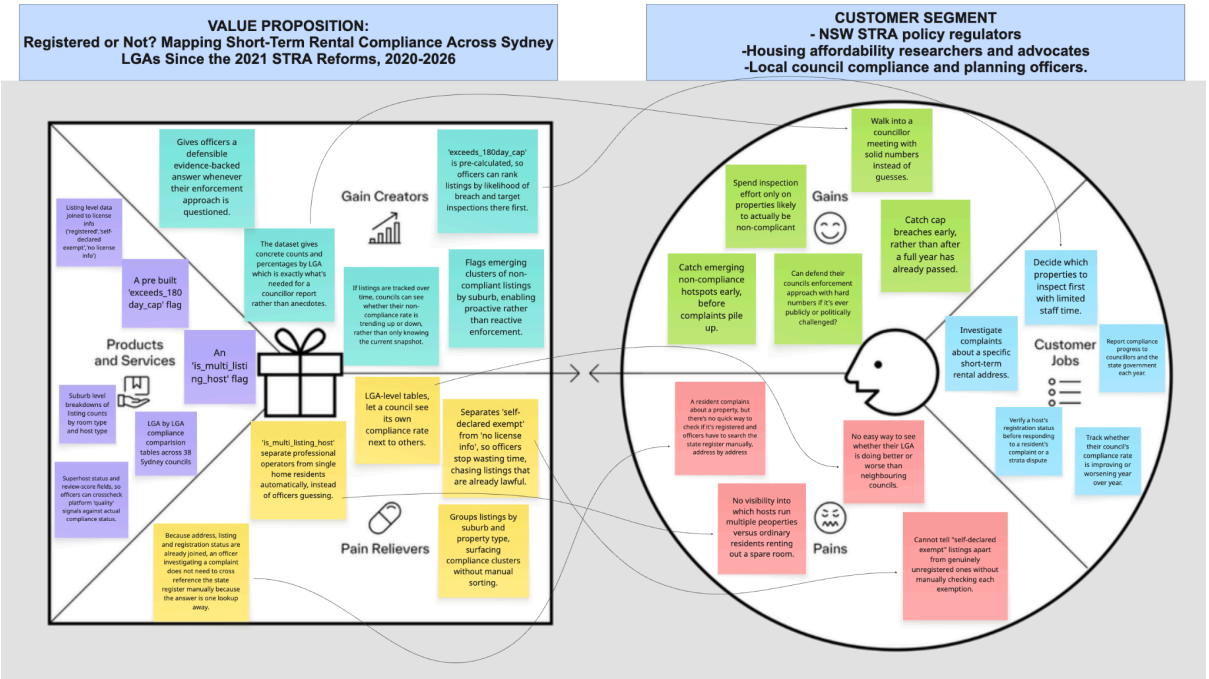
Unique Contribution

Rather than simply counting listings, this project links room type, host scale, registration status and availability to identify listings that behave like full time rental businesses. We distinguish formally registered properties from self declared "Exempt" listings and those with a genuine PID-STRA number, since claiming exemption is not the same as being registered. This gives a clearer picture than registration data alone.

Broader Application

Councils could use this analysis to prioritise investigations rather than relying solely on missing registration numbers. Combined with vacancy rates, median rents, housing supply and population data, it could reveal whether higher Airbnb concentration drives stronger rent growth, particularly in the City of Sydney, Waverley, Randwick and the Northern Beaches.

The Value Proposition Canvas:

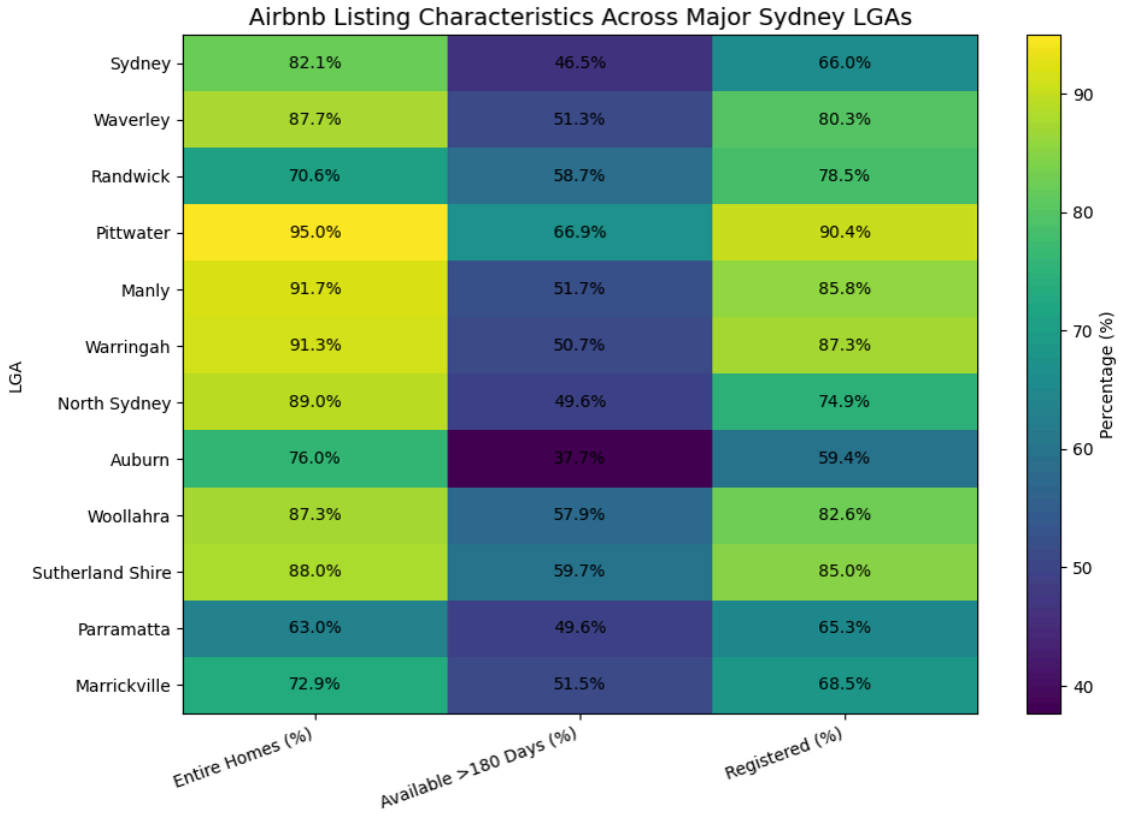


Key Insights from the Data

- In the 17,741 home/apartments listed, the entire home/apartments make up around 77.8% (13,805).
- There is a high degree of extended short-term rental availability as roughly 48% (6,663) of all listed homes are available for more than 180 days a year.
- The median availability of listings with no licence information is around 357 days, while the registered listings have a median availability of 187 days.

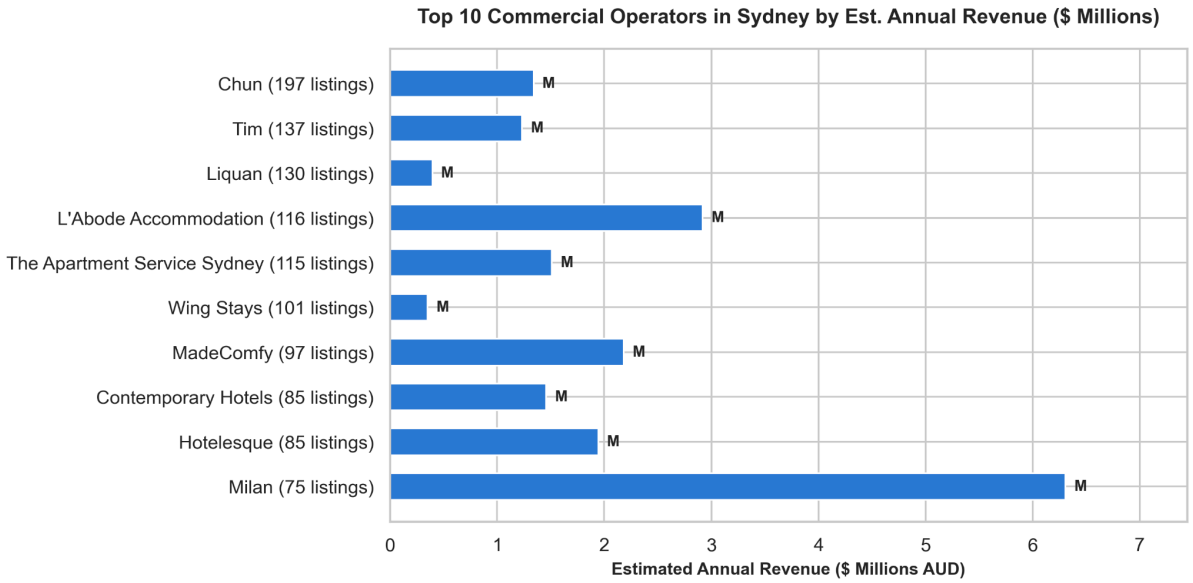
- Airbnb activity is also geographically localised, with the City of Sydney having about 24% of all Airbnb properties.

	lga	Total_Listings	Entire_Home_Pct	Over_180_Days_Pct	Registered_Pct
30	Sydney	4307	82.1	46.5	66.0
33	Waverley	1479	87.7	51.3	80.3
25	Randwick	908	70.6	58.7	78.5
24	Pittwater	794	95.0	66.9	90.4
18	Manly	745	91.7	51.7	85.8
32	Warringah	732	91.3	50.7	87.3
21	North Sydney	716	89.0	49.6	74.9
0	Auburn	621	76.0	37.7	59.4
35	Woollahra	511	87.3	57.9	82.6
29	Sutherland Shire	459	88.0	59.7	85.0
22	Parramatta	427	63.0	49.6	65.3
19	Marrickville	410	72.9	51.5	68.5

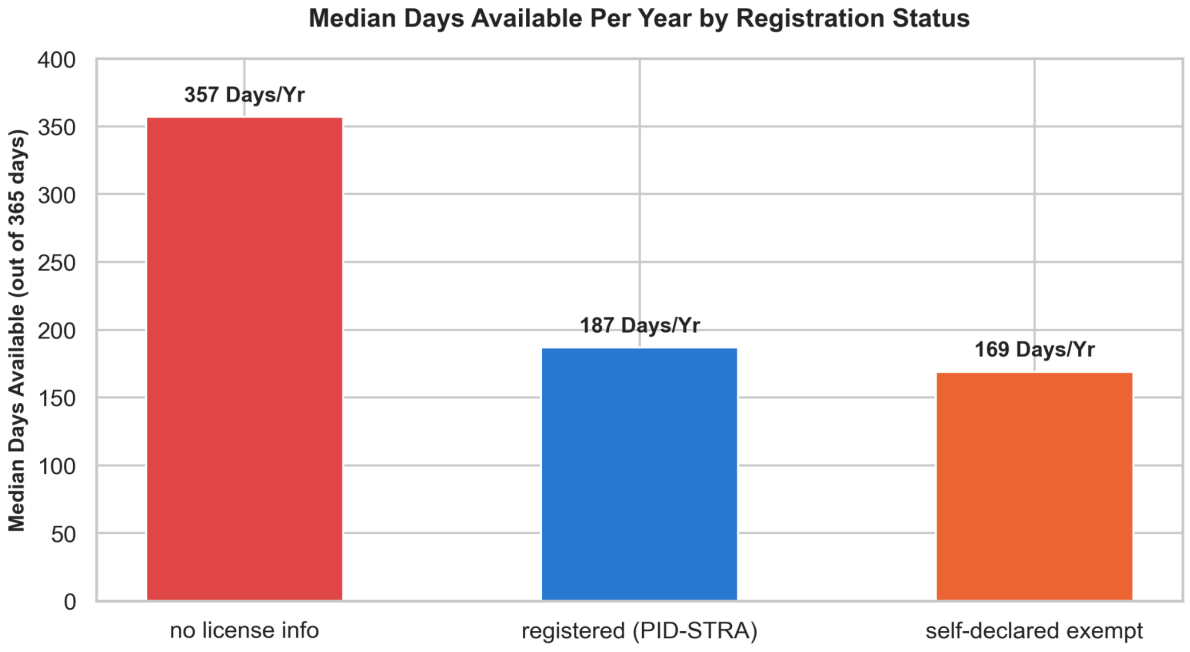


One limitation is that the dataset represents a snapshot of Sydney Airbnb listings at a specific point in time, so it cannot capture how listing patterns and host behaviour have changed over time.

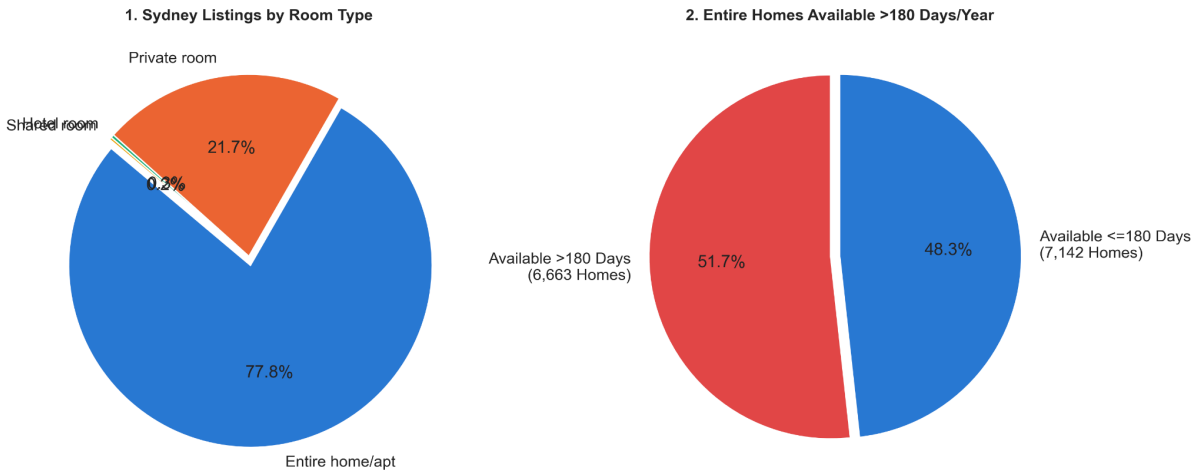
The initial patterns indicate that Airbnb is more than just an occasional sharing of homes in Sydney. This will be further analysed to determine the interaction between location, registration and host type and availability to find areas where short-term rental activity is concentrated.



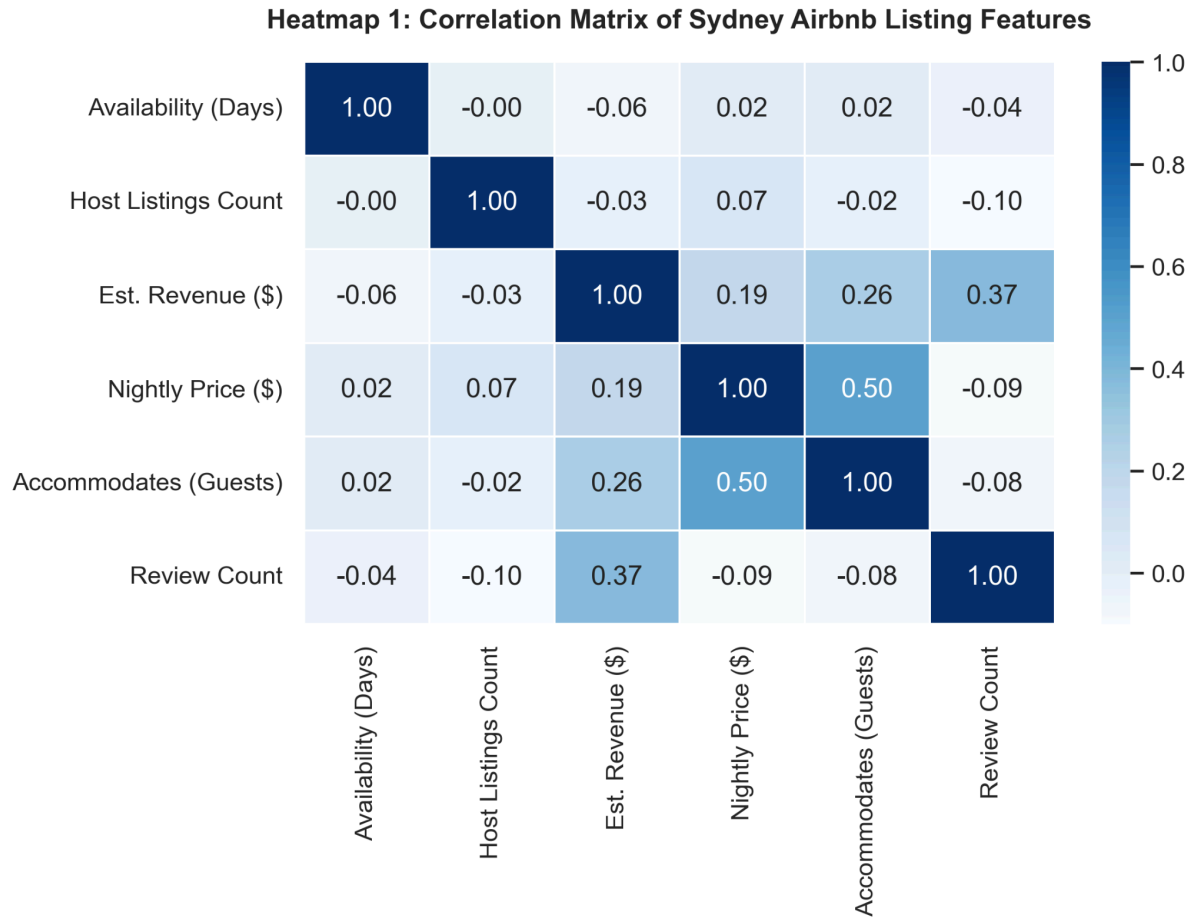
Commercial Concentration: Top operators hold up to 209 listings, generating $2.9M– $6.3M annually.



Compliance Disparity: Unregistered listings average 357 days median availability vs. 187 days for registered stock.



Housing Stock Drain: 77.8% of listings are whole homes, with 48.3% available >180 days/year.



Operational Dynamics: Commercial host scale positively correlates with year-round listing availability.

References

- City of Sydney. (2025, October 27). *Capping short-term rental accommodation in the City of Sydney*. https://meetings.cityofsydney.nsw.gov.au/mgDecisionDetails.aspx?IId=34530&Opt=1
- Gurran, N., Buckle, C., Troy, L., Nasreen, Z., Crommelin, L., Phibbs, P., Shrestha, P., Redmond, D., & Harrison, J. (2026). *Short-term rental accommodation: Models, impacts and policy responses* (AHURI Final Report No. 460). Australian Housing and Urban Research Institute. https://doi.org/10.18408/ahuri7333501
- Inside Airbnb. (2026). *Sydney, New South Wales, Australia: 16 June 2026* [Data set]. Retrieved August 18, 2026, from https://insideairbnb.com/get-the-data/